

# Detecting Factual Contradictions in News Based on Large Language Models

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**Abstract:** Online news provides readers with abundant information sources, yet the authenticity of such information is often not guaranteed. Determining the veracity of news has become an important research topic in recent years. Prior work has achieved considerable success in identifying deliberately fabricated information, but existing methods struggle to handle reports that were not originally intended as fabrications yet were subsequently proved false. This paper proposes a method that detects factual contradictions in news by cross-validating reports describing the same event, since contradictory facts are a prerequisite for determining veracity. The method consists of two main tasks: identifying news reports that express a denial, and identifying the corresponding denied reports; the two together constitute a contradictory news pair. We also release CNCD (Chinese News Contradiction Dataset), a Chinese news dataset dedicated to analyzing contradictions in news veracity, which contains 5,022 contradictory news pairs. Experiments on CNCD show that the method in the first task achieves a recall rate exceeding 97%, and that the method in the second task can effectively identify mutually contradictory news pairs within a news collection.

**Keywords:** Fake news detection; Fact extraction; Large language model

## 1 Introduction

In the current era of information explosion, the Internet provides people with rapid access to information, but the authenticity of online information is difficult to guarantee. In recent years, a substantial body of research has been devoted to the identification of false information; these studies judge whether a piece of news contains fabricated content on the basis of the given news text, images, comments, or propagation paths.

For a long time, fake news detection has largely focused on information samples produced with the intent to deceive. Because the authors of such messages deliberately set out to fabricate content, the resulting messages often exhibit features that distinguish them from genuine information, such as unusual symbols, inflammatory language, and inconsistencies between text and images; these features can be exploited for detection. The propagation paths of such fabricated information on social networks also differ from those of authentic information, which provides another discriminative feature.

It is worth noting, however, that one situation has long been overlooked. Suppose a news report (hereafter referred to as News A) is published; its author believes that the information

was obtained from reliable sources and supports it with corroborating material. News A contains no contradictory or fabricated content and may therefore, for the time being, be regarded as true. As time passes, a party mentioned in News A issues a denial of the information it contains. Another report (News B) covers this denial statement, and consequently News A and News B come into conflict. We refer to News A as a denied report and to News B as a denial report; News A and News B together constitute a contradictory news pair, and one of the two reports in such a pair must be false. However, analyzing the denied report or the denial report in isolation does not allow one to determine its veracity effectively. Because the authors of both reports believe that what they describe is factual, and because both wrote with the intention of disseminating true information, such reports typically exhibit neither the intrinsic features of false information nor distinctive propagation patterns. Existing fake news detection methods therefore struggle to handle this category of news.

Investigating such contradictions in news veracity is of great importance for tracking the evolution of news events. It should be emphasized that, although News B denies the content of News A, this alone does not imply that the content of News A has become false. In the absence of reliable external evidence, it is sometimes difficult to determine which report in a contradictory news pair is the false one. Therefore, the present work focuses on detecting and discovering contradictory news pairs.

This paper divides the detection of contradictory news pairs into two tasks. Task 1 identifies denial reports: candidate news is first screened by a filter based on lexical features, and a large language model (LLM) is then employed to further determine whether the news denies certain information and to extract the denied content. Task 2 locates the denied reports corresponding to the denial reports: the denied content extracted in Task 1 is used to retrieve from the news collection, and multiple filters are applied to the retrieved news to make the final judgement. News that passes the judgement becomes a denied report and forms a contradictory news pair together with the denial report.

We also release CNCD (Chinese News Contradiction Dataset), a Chinese news dataset for analyzing contradictions in news veracity. The dataset contains 105,361 Chinese news articles collected from public news websites, together with the title, body text, and publication date of each article. The IDs of the denial reports are provided, amounting to 1,957 articles, as well as the news ID pairs of contradictory news pairs, amounting to 5,022 pairs. Figure 1 presents an example of a denial report and its corresponding denied report.

News 1136500 (news ID) is a denial report, in which the denied information is shown in italic text. News 1115112 (news ID) is the denied report corresponding to News 1136500; 1136500 and 1115112 form a contradictory news pair.

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| Denial Report                                                                                                                                                                                                                                           |    | Denied Report                                                                                                                                                                        |

**Fig. 1 An example of a contradictory news pair**

We hope that this work will advance the state of the art in detecting contradictions in news, thereby enabling better tracking of how reported events evolve over time and improving the detection of potential misinformation.

## 2 Related Work

Fake news detection has emerged as an active research topic in recent years, and a substantial body of work has been devoted to it. Early studies relied heavily on manually designed features. For example, Liu et al. [1] proposed to determine veracity by checking whether a message contains eyewitness phrases such as “I see” and “I hear”, on the grounds that the presence of such words makes a news message appear more credible. Biyani et al. [2] extracted features from URLs, such as the frequency of hyphens, capital letters, and commas. Gupta et al. [3] used the occurrence counts of specific words as features, for instance the usage of first-, second-, and third-person pronouns. Jin et al. [4] proposed a three-layer hierarchical model comprising an event layer, a sub-event layer, and a message layer, and employed a graph-based iterative optimization algorithm to compute the final event credibility.

Ma et al. [5] exploited the MPQA3 sentiment lexicon together with a set of manually collected high-frequency emoticons to identify positive or negative words, with the aim of using sentiment features to detect inflammatory statements containing emotional words or sentences. Hassan et al. [7] likewise used sentiment-bearing words as features, but employed the Natural Language Toolkit (NLTK) to tag the sentiment feature words.

Several studies have found that the propagation patterns of false messages differ from those of true ones, and that fake news can therefore be identified by analyzing propagation patterns. Wu et al. [9] proposed computing forwarding-time features, that is, the average time difference between an original message and its reposts, reasoning that spammers tend to repeatedly repost and comment on similar content. Ma et al. [10] observed that fake news is usually first released by users with low influence and then widely disseminated by opinion leaders, whereas true news is generally initiated by opinion leaders and subsequently propagated by ordinary users.

The advance of deep learning has given rise to numerous methods that extract features with deep neural networks and then analyze propagation patterns. Ma et al. [11] proposed the use of recurrent neural networks for fake news identification; this model is able to learn long-range dependencies within a message. Ruchansky et al. [12] proposed a hybrid deep model that combines textual content, user responses (i.e., reposts and comments), and source user information, with the aim of achieving more accurate fake news detection.

In recent years, the remarkable performance of large language models has motivated research on using them for fake news identification. For instance, the studies of Caramancion [13][14] show that, although LLMs still fall short of human fact-checkers in distinguishing true from false news, they have already attained a moderate level of ability and exhibit considerable potential.

### **3 The Proposed Method**

The focus of this work is on discovering contradictions between two reports, and contradictions take time to emerge. After an event is reported, if it attracts a certain degree of attention, follow-up reports will appear. These follow-up reports may contain information that differs from the initial report; typically, a participant or an interested party issues a denial statement, or an authoritative body (such as the police or a government department) publishes the findings of an investigation. Such follow-up reports containing denial content are the denial reports referred to above. Accordingly, this work starts by detecting denial reports: at the current point in time, denial reports are detected from the news stream; when a denial report is found, historical news is traced back to find reports describing the same event within a certain period; from these historical reports, the ones corresponding to the content denied in the denial report—that is, the denied reports—are selected. The two together form a contradictory news pair. The proposed algorithm therefore consists of two tasks: Task 1 detects denial reports, and Task 2 locates the denied reports.

### 3.1 Task One

The task of Task 1 is formally described as follows: given the ID, title, and body text of a news article, determine whether the article explicitly denies certain information mentioned in it. Figure 2 presents a contrastive example of an explicit denial and a non-denial. News 3408757 explicitly denies the report that 20 Wanda Plazas were sold, whereas News 3389696 expresses no opinion on that information.

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|------------------------------------------------------------------------------------------------------------------------------------------------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><div>ID:3408757</div><div>TITLE:万达集团声明:网传“160亿销售20个万达广场”为不实消息<br/>CONTENT:乐居财经 李礼 5月25日,万达集团官网披露集团声明,称网传“160亿销售20个万达广场”为不实消息。</div></div> | VS | <div><div>ID:3389696</div><div>TITLE:卖20家购物中心回笼现金?万达回应“不予置评”<br/>CONTENT:据界面新闻,5月24日,市场消息称,万达集团正在考虑评估出售上海、江苏、浙江等地的20家购物中心,每个项目寻求约7亿元至8亿元的估值,具体取决于位置和业务,上海一家购物中心的售价可能高达10亿元。对此,万达集团回应称“不予置评”。.....</div></div> |
| Explicit Denial                                                                                                                                |    | No Explicit Denial                                                                                                                                                                                            |

Fig. 2 Examples of an explicit denial and no explicit denial

Interestingly, chained denials sometimes occur: a news report denies a certain piece of information, and some time later a new report denies the previous one. Figure 3 gives such an example.

|                                                                                                                                                                                                                                   |   |                                                                                                                                                                                                                              |
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| <div><div>ID:3315885</div><div>TITLE:万达辟谣:网传万达大规模裁员消息不实 全球聚看点<br/>CONTENT:(资料图片仅供参考)\n2023年5月20日,万达集团官网发布声明称:网传万达大规模裁员消息不实。近期,坊间传闻,“万达大规模裁员”“被曝出大规模裁员的消息”“进行大规模裁员,涉及比例达30%以上”,对于这些消息,万达集团官网用了简短的“不实”来进行回复。封面图片来源:万达官微</div></div> | ⇒ | <div><div>ID:3342936</div><div>TITLE:社区平台,万达员工辟谣了万达辟谣!<br/>CONTENT:.....不过,万达近期对于裁员30%进行辟谣,万达声明:网传万达大规模裁员消息不实。不过,对于万达这份声明,广大网友和供应商们明显表示不服。有不少万达的员工们表示,不知道万达有啥好辟谣的,现实就摆在那里。而且万达的这份辟谣,只是辟谣了大规模裁员不实,并未辟谣裁员这件事。.....</div></div> |
| Denial                                                                                                                                                                                                                            |   | Denial Chain                                                                                                                                                                                                                 |

Fig. 3 An example of a denial chain

It must be pointed out that, even if a news report explicitly denies a certain piece of information, the denied information cannot be taken as necessarily false. A number of complications arise: a party involved in the event may be temporarily unable to acknowledge its occurrence for reasons of confidentiality or otherwise; or a party may deny having done something out of self-protection. It can also be observed in some examples that the public or other media outlets do not accept the “denial” issued by certain parties. See, for instance, News 2781346 (Appendix 1).

To identify denial reports, we first construct a matching filter that uses a matching-based method to rapidly discard news that does not contain negation semantics, and then employ an LLM for further analysis.

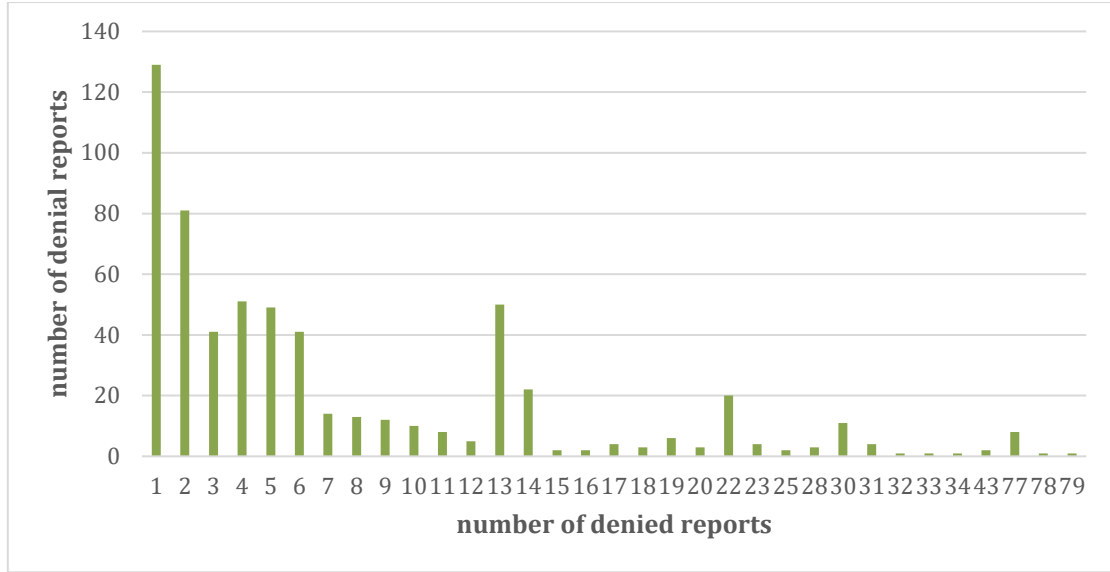
The matching-based method comprises positive feature matching and negative feature matching. A positive feature is text that can or may express the meaning of “denial”, whereas a negative feature is text in which a positive feature appears but whose semantics do not in fact express denial. For example, “澄清” (to clarify) is a positive feature: in “Company A issued a clarification regarding the online rumor”, the word indicates that a further explanation of the rumor will be given, which may involve a denial. In contrast, in a phrase such as “河道澄清剂 3 月 23 日报价” (quotation for a river clarification agent on March 23), the same string should not be matched as a positive feature.

We construct a positive feature set and a negative feature set. Given an input news article, its title and first sentence (if present) are taken as the text to be matched. The negative feature set is applied first, replacing any negative feature strings in the text with non-word characters. The positive feature set is then applied; if the text matches a positive feature, the article proceeds to the next stage.

We then use an LLM to determine whether the news article explicitly denies something that has already been disclosed. If the model answers “yes”, the article is classified as a denial report, and the model is additionally asked to extract the denied event together with its main subjects and locations. The specific prompts and examples are given in Appendix 2.

### **3.2 Task Two**

The task of Task 2 is described as follows: given a set of denial reports and a news collection, find, from the news collection, the denied reports corresponding to the news in the denial report set, so as to form a set of contradictory news pairs. A denied report is one that reports the information denied in a denial report. A denied report must not itself deny that information, and its publication time must be earlier than that of the denial report. One denial report and one corresponding denied report constitute a contradictory news pair. A denial report may correspond to multiple denied reports, thereby forming several contradictory news pairs, or it may have no corresponding denied report at all. Figure 4 shows the distribution of denial reports according to the number of denied reports they correspond to; the horizontal axis denotes the number of denied reports corresponding to a single denial report, and the vertical axis denotes the number of denial reports having the same number of denied reports.



**Fig. 4 Distribution of denial reports by their number of denied reports**

We first retrieve from the news repository using the content of the denied event extracted in Task 1, and then filter the retrieved news so as to ensure that the retained news is consistent with the content of the denied event.

### 3.2.1 Retrieval

We experiment with a vector database (Faiss [18]), a discrete search engine (Elasticsearch), and a combination of the two. When using Elasticsearch, the titles and body texts of all news articles in the dataset are first added to the index; at retrieval time, the text of the denied message extracted in the previous step is used as the query, and the maximum number of returned candidate articles is set to 100. When using Faiss, we note that news articles are usually long, and that embedding the entire body text may not adequately reflect the main content. We therefore concatenate the title with the first paragraph of the body whose length exceeds 20 characters, embed the resulting text, and add the vector to the index. At retrieval time, the denied message text is embedded into a vector, which is then used for retrieval; the maximum number of returned candidate articles is set to 50. The text embedding model used in this work is BGE-3 [19, 20, 21]. If Elasticsearch and Faiss are used simultaneously, the news returned by both is merged by news ID to obtain the candidate news set.

### 3.2.2 Filtering

First, news whose publication time is later than that of the denial report is removed from the retrieval results. Filters are then applied to the retrieved news set (referred to as the

candidate news set), and the news that survives is the denied report corresponding to the denial report. Three filters are used: the LLM Filter, the ENT Filter, and the SIM Filter.

ENT Filter: this filter checks whether all the relevant subjects and locations extracted above appear in a retrieved news article; only articles containing all of them pass this filter and proceed to the next stage.

SIM Filter: this filter examines the textual similarity between a candidate news article and the denial report. The underlying assumption is that if two news articles describe the same event but express different views on its veracity, they should exhibit a relatively high textual similarity, though not an excessively high one. Based on our experiments, we adopt the interval  $[0.6, 0.8]$ . Candidate news whose similarity to the denial report falls within this interval passes the SIM Filter and proceeds to the next stage. As before, the text used for vectorization is the concatenation of the title and the first paragraph of the body whose length exceeds 20 characters.

LLM Filter: an LLM is used to judge the relation between a candidate news article and the denial report. We design two prompts. The first prompt checks event consistency, that is, whether the two articles describe the same event: the model is asked first to summarize the main content of each of the two input texts in one sentence, and then to judge whether the two summaries concern the same event. The second prompt checks content exclusivity, asking the model to judge whether the main contents of the two input texts are in conflict. The specific prompts are given in Appendix 3 and Appendix 4. The text fed to the model is not the full text of the news article. Because a single news article may describe multiple events, we design an algorithm that rapidly extracts from an article the passages relevant to the denied message; the extracted passages are then fed to the model.

The article is first split into  $N$  paragraphs along natural paragraph boundaries, and a score is computed for each paragraph.

*Let  $E_i$  denote the set of distinct event-related entities (relevant subjects and locations) appearing in the  $i$ -th paragraph, and let  $Score_i$  denote the score of the  $i$ -th paragraph, i.e., the number of distinct relevant entities occurring in paragraph  $i$ :*

$$Score_i = |E_i| \quad i \in [1, N]$$

The scores of all paragraphs are then examined. Let  $m$  and  $n$  be, respectively, the smallest and the largest index among the paragraphs attaining the highest score; the paragraphs from  $m$  to  $n$  are concatenated to form the finally selected text. When  $m = n$ , the selected text consists of a single paragraph. The algorithm can be formalized as follows:

$$\begin{aligned}
m &= \min\{i | \text{Score}_i = \max(\text{Score}_j)\} j \in [1, N] \\
n &= \max\{i | \text{Score}_i = \max(\text{Score}_j)\} j \in [1, N] \\
T &= P_m \oplus P_{m+1} \oplus \dots \oplus P_n
\end{aligned}$$

where  $P_m$  is the first and  $P_n$  the last of the highest-scoring paragraphs,  $T$  is the finally selected text, and  $\oplus$  denotes the concatenation of all paragraphs from  $P_m$  to  $P_n$ .

The LLM Filter is the final and mandatory filtering step, whereas the ENT Filter and the SIM Filter are optional. The experiments reported below evaluate the effect of different filter combinations.

## 4 Dataset Construction

This section describes the construction of CNCD, including data collection, the annotation of denial reports, the annotation of contradictory news pairs, and the addition of negative samples.

### 4.1 Data Collection

We collected news articles published between March 2023 and May 2023 from dozens of public Chinese news websites, amounting to 2,172,523 articles in total. The news is mainly concentrated in the economic domain. We take economic news as the starting point of CNCD, with the aim of enabling effective research on news contradictions; we plan to extend the dataset to other domains in future work.

### 4.2 Finding Denial Reports

Compared with the total number of collected articles, the number of denial reports is relatively small, and manually searching for denial reports among millions of articles is difficult and inefficient. We therefore adopt a heuristic approach. Because a denial report explicitly expresses a negative stance toward certain content, it must contain words conveying negation. We collect 20 negation features and combine them into a regular expression. This regular expression is matched against the title and the first sentence of each collected article, and the articles that match are retained. Human experts then manually inspect the retained articles to determine whether they are denial reports. In the end, 1,957 denial reports are obtained. The information of the denial reports is recorded in JSON format in “CNCD.jsonl.task1.positive”, including the news ID, title, body text, and publication date.

### 4.3 Finding the Corresponding Denied Reports

For each denial report, we use an LLM (DeepSeek-V3 [22]) to summarize three keywords describing the denied content and the parties involved, and then use these

keywords to retrieve from news whose publication time is no later than that of the denial report. Human experts then judge whether the retrieved news can form a contradictory news pair with the denial report.

Not every denial report has a corresponding denied report. The content denied in some denial reports originates from social media, short-video platforms, or other media, so no corresponding content can be found in the news. Of the 1,957 denial reports, 600 were matched with corresponding denied reports, yielding 5,022 contradictory news pairs in total. Detailed statistics of the news counts are given in Table 1. The data of the contradictory news pairs are recorded in “CNCD.jsonl.task2.CRP”.

**Table 1 Distribution of label types**

| Label Type                                       | Count |
|--------------------------------------------------|-------|
| Denial reports                                   | 1,957 |
| Denial reports with corresponding denied reports | 600   |
| Denied reports                                   | 800   |
| Contradictory news pairs                         | 5,022 |

#### 4.4 Adding Negative Samples

The denial reports and their corresponding denied reports obtained by the above procedure constitute the positive samples. In order to evaluate the effectiveness of the algorithm, noise samples also need to be added to the dataset. We augment the dataset with noise samples in two ways. (1) When searching for denied reports, the retrieved news, after removing the positive samples, is taken as the noise sample set N1. (2) After removing the positive sample set and the noise sample set N1 from all the collected news, we randomly sample 50,000 articles from the remainder as the noise sample set N2. N1 and N2 are merged into the noise sample set. Merging the labeled sample set and the noise sample set yields the CNCD news collection, which contains 105,361 news articles in total. The DOI of the dataset is 10.57760/sciencedb.36026.

## 5 Experiments

### 5.1 Experiments on Task 1

All news articles in CNCD are fed into the matching filter, and 2,439 articles are retained after filtering. The retained articles are then fed into the LLM (DeepSeek-V3) to determine whether they are denial reports. Ultimately, 2,172 articles are identified as denial reports. The evaluation metrics of Task 1 are the precision, recall, and F1-score of denial report identification; the experimental results are shown in Table 2.

**Table 2 Performance metrics of Task 1**

| Method | Precision | Recall | F1-score |
|--------|-----------|--------|----------|
| Task 1 | 88.22     | 97.91  | 92.81    |

As the results show, the recall is relatively high while the precision is somewhat lower. We analyzed the erroneous samples and found that the errors mainly fall into the following categories:

- The news denies certain matters, but the article does not explicitly state the original information that is being denied, as shown in Appendix 5.
- The news provides an explanation of certain matters, which does not amount to a denial, as shown in Appendix 6.

## 5.2 Experiments on Task 2

The evaluation metrics of Task 2 are the precision, recall, and F1-score of the finally matched contradictory news pairs. We conduct a filter combination experiment and a retrieval strategy combination experiment.

### 5.2.1 Filter Combinations

In this experiment, four filter combinations are tested: ENT Filter + SIM Filter + LLM Filter, ENT Filter + LLM Filter, SIM Filter + LLM Filter, and LLM Filter only. The filters are applied in the same order as they appear in each combination. The precision, recall, F1-score, and running time of each combination are reported in Table 3.

**Table 3 Results of different filter combinations**

| Filter Combination                   | Precision    | Recall       | F1-score     | Time           |
|--------------------------------------|--------------|--------------|--------------|----------------|
| ENT Filter + SIM Filter + LLM Filter | <b>67.53</b> | 24.74        | 36.21        | <b>17.63 h</b> |
| ENT Filter + LLM Filter              | 63.71        | 47.18        | 54.21        | 60.45 h        |
| SIM Filter + LLM Filter              | 61.02        | 35.07        | 44.54        | 101.73 h       |
| LLM Filter                           | 59.37        | <b>67.08</b> | <b>62.99</b> | 170.99 h       |

As can be seen from the results, using the LLM Filter alone yields an F1-score substantially higher than those of the other combinations, but at the cost of a greatly increased running time. Using all the filters gives the highest precision and the lowest running time. This indicates that the ENT Filter and the SIM Filter remove a large number of news articles: while greatly improving computational efficiency, they also discard many correct results. There is thus considerable room for improving these two filters.

### 5.2.2 Retrieval Combinations

In this experiment, three retrieval strategies are tested: Elasticsearch alone, Faiss alone, and Elasticsearch and Faiss combined. The experiment is conducted under the setting where only the LLM Filter is used. Table 4 reports the final metrics of the three retrieval strategies.

**Table 4 Results of different retrieval strategies**

| <b>Retrieval Strategy</b> | <b>Precision</b> | <b>Recall</b> | <b>F1-score</b> | <b>Results Count</b> |
|---------------------------|------------------|---------------|-----------------|----------------------|
| Elasticsearch             | 58.69            | 60.51         | 59.59           | 5,178                |
| Faiss                     | 67.15            | 41.55         | 51.34           | 3,108                |
| Elasticsearch + Faiss     | 59.37            | 67.63         | 63.23           | 5,721                |

As shown in Table 4, Elasticsearch and Faiss contribute to the results in different ways: Elasticsearch recalls a larger number of results, whereas Faiss yields more precise results. Not listed in Table 4 is that the intersection of the results of the two retrieval strategies contains 2,565 items, of which 1,727 are correct; beyond this intersection, both Elasticsearch and Faiss each contribute a considerable number of additional correct results. It follows that, if high precision is required, Faiss alone should be used; otherwise, using Elasticsearch and Faiss together is the best choice.

The experiments demonstrate that the proposed method can effectively identify contradictory news pairs, and they also reveal several directions for further improving the task metrics.

- In Task 1, the semantics of the news should be analyzed in greater detail, for example by distinguishing “explanation” from “denial”, and by verifying whether the article does in fact quote the original statement of the denied content.
- In Task 2, two text truncation methods are used: one truncates the title and the leading part of the article when embedding the news, and the other extracts the passages relevant to the denied message when filtering with the LLM. Both methods leave room for improvement.
- Some news articles describe multiple events within a single article; for such articles, the contents of the individual events should be separated and processed independently.
- In Task 2, some news articles narrate the original information corresponding to a denial report but subsequently deny it. Identifying such articles would help improve the metrics.

## 6 Conclusion

Targeting the problem that reports of the same event may contradict each other across different news articles, this paper designs an effective method for identifying contradictory news pairs. The method identifies denial reports and their corresponding denied reports in two tasks. Experiments show that the method can effectively identify contradictory news pairs, although room for improvement remains. We also construct CNCD, a Chinese news contradiction dataset containing denial reports and the corresponding denied reports. This

dataset is expected to advance research on detecting contradictions in news and on tracking the evolution of news events.

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## Appendix

### 1. The public and the media don't accept such denials.

{"id": 2781346, "date": "2023-05-02", "title": "多家出版机构解除合作，编剧史航回应性骚扰指控：情绪我理解，但情况不属实", "content": "文/羊城晚报全媒体记者 李丽\n图/微博@鹦鹉史航、@磨铁图书、@新周刊\n5月1日，编剧史航在微博回应近日的性骚扰指控：“网上涉及我的文字，我看到了，情绪我理解，但情况不属实。这三天，我一直没有回复网络上的诸多质疑，因为我正试图找到妥当的办法，在保护双方隐私的情况下，还原事实。然而，给自己相识的女性带来如此的感受，是我真正的失败，十分歉疚。”\n此前数日，曾有多名女性通过各种途径反映，史航曾在过去对她们进行性骚扰。这是史航第一次就此事进行回应。\n与此同时，已有多家出版机构发公告称，与史航解除相关合作。\n对于史航的回应，网友们在评论区发表看法：“总结：我连狡辩的词都想不出。”“总结：别急，我还没编好。”“有这么多‘相识’的女性能指出共同的‘感受’，你不会要说这是你表达礼节的方式吧。””}

### 2. Prompt template for task1.

## 2.1 Prompt Template for Denial Detection and Denial Content Extraction

message = [

    {"role": "system",

    "content": "你是一个擅长阅读理解新闻的专家。阅读给出的新闻内容然后回答：\n

    根据给出的新闻回答新闻的主要内容是否针对某些已经披露的事情进行了明确地

    否认，以“是”或“否”进行回答；"},

    {"role": "user",

    "content": "特斯拉要在沙特建厂？马斯克在线辟谣\n9月18日，《华尔街日报》

    援引知情人士的消息报道称，沙特阿拉伯正与美国电动汽车制造商特斯拉就在沙

    特建立制造工厂进行初步谈判。报道称，沙特阿拉伯一直在向特斯拉抛出橄榄枝，

    试图用从刚果民主共和国等国购买特斯拉电动汽车所需的一定数量的金属和矿产

    的权利来吸引其到当地建厂。据悉，沙特正在考虑的一项举措是向大宗商品交易

    商 Trafigura 提供融资，帮助其完成刚果的一个钴铜项目，该项目可能有助于为特

    斯拉的一家工厂提供供应。此前，在成本不断上升和钴价持续低迷的情况下，

    Trafigura 正在评估其对刚果 Mutoshi 项目的选择。沙特一直在努力使其经济摆脱

    对石油的依赖，除了尝试吸引特斯拉之外，该国主权财富基金还是美国电动汽车

    初创公司 Lucid Group 的主要投资者。针对上述报道，特斯拉首席执行官埃隆·马

    斯克在社交媒体平台 X 上发帖进行了否认，而沙特阿拉伯的公共投资基金

    （Public Investment fund）拒绝置评。特斯拉目前在全球拥有 6 家工厂，并正在墨

    西哥北部新莱昂州建设第七家工厂。今年 5 月，马斯克曾表示，特斯拉可能会在

    今年年底前敲定下一个工厂的最终选址。今年 8 月，特斯拉表示有兴趣在印度建

    立一家生产低成本电动汽车的工厂。日前，外媒又报道称土耳其总统雷杰普·塔伊

    普·埃尔多安（Recep Tayyip Erdogan）在美国纽约与马斯克会面，并邀请特斯拉在

土耳其设立工厂。9月18日，马斯克还在加州会见了以色列总理 Benjamin Netanyahu。特斯拉的目标是到2030年每年销售2000万辆汽车，较2022年的130万辆将翻数倍。"}，

```
{"role": "assistant",  
  "content": "是"},  
  
{"role": "user",  
  "content": title + '\n' + content}
```

]

## 2.2 Prompt Template for JSON Conversion:

```
message = [  
  {"role": "system",  
    "content": "将下面的文本整合为如下的 json 格式：{'has_rumor': True/False,  
  '\rumor': [{'subject': [被否认事情相关的主要主体], 'number': [重要数据],  
  '\location': [主要地点], '\rumor_content': 否认的事情}]}。"},  
  {"role": "user",  
    "content": '1. 文中有涉及到对某些事情进行了明确地否认。 \n2. 被否认的事情是："该等公司为国美非重要附属公司，冻结该等公司股权不会对公司的财务状况和经营产生重大影响。" \n3. 主要主体：国美零售控股有限公司；主要地点：空；重要数据：68.9 亿元。'},  
  {"role": "assistant",  
    "content": '{\n  "has_rumor": true,\n  "rumor": [\n    {\n      "subject": ["国美零售控股有限公司"],\n      "number": [68.9 亿],\n      "location": [],\n      "rumor_content": "该等公司为国美非重要附属公司，冻结该等公司股权不会对公司的财务状况和经营产生重大影响。"\n    }\n  ]\n}',  
  {"role": "user",  
    "content": answer_of_extraction}  
]
```

## 2.3 Example: news 1099088 and its Extraction Result

News 1099088:

```
{  
  "id": 1099088,  
  "date": "2023-03-03",  
  "title": "网传“响应中国共富工程 80000 套安置房分配方案”系虚假文件",  
  "content": "3月3日消息：近期，陆续接到群众反映，咨询北
```

京市住房和城乡建设委员会是否发布过《关于印发&响应中国共富工程 80000 套安置房分配方案的通知》(京建发〔2023〕28 号)。经核实,我委未制作印发过《关于印发&响应中国共富工程 80000 套安置房分配方案的通知》(京建发〔2023〕28 号)。不法分子涉嫌伪造国家机关公文,我委已向公安机关报案。请广大群众提高警惕,加强甄别,谨防上当受骗!"}

Extraction Result:

```
{"rumor_data": {"has_rumor": true, "rumor": [{"subject": ["北京市住房和城乡建设委员会"], "number": [80000], "location": ["北京市"], "rumor_content": "北京市住房和城乡建设委员会发布过《关于印发&响应中国共富工程 80000 套安置房分配方案的通知》(京建发〔2023〕28 号)"}]}}
```

### **3. Prompt template for verifying whether the candidate article and the Denial Report describe the same event, in task2.**

请阅读以下两个文本,分别各用一句话总结两个文本的主要内容,并判断两个文本的主要内容是否涉及了同一件事情,以“是、否”做出回答。文本 1: 【%s】。文本 2: 【%s】。

### **4. Prompt template for judging whether there is a conflict between the candidate article and the Denial Report, in task2.**

请阅读以下两个文本并判断文本 1 的主要内容是否全面或部分地否定了文本 2 的主要内容,以“是、否”做出回答。文本 1: 【%s】。文本 2: 【%s】。

### **5. The news denies certain events, but the original reports of these events are not explicitly stated in the text.**

```
{"id": 1320329, "title": "永鼎股份澄清: 公司不涉及“室温超导”相关业务", "content": "e  
公司讯, 永鼎股份(600105)3 月 9 日晚间发布澄清公告, 公司主营业务为通信科技、海  
外电力工程、汽车线束及超导电力, 不涉及“室温超导”相关业务, 也未开展相关研发
```

和投入。公司下属全资子公司东部超导科技苏州）有限公司主营第二代高温超导带材及其应用设备以及超导（通用）电气产品，与“室温超导”无关。”}

**6. The news provides explanations or clarifications regarding certain events, but such content cannot be regarded as a Denial Report.**

{"id": 1321073, "title": "舍得酒业澄清：公司内外部生产经营环境未发生重大变化 内部生产秩序及相关经营情况一切正常", "content": "舍得酒业（600702）(600702.SH)发布澄清说明公告，近日，公司关注到相关媒体对公司生产经营及相关事项的报道信息，为避免对投资者产生误导，现予以澄清说明：一、经自查，公司内外部生产经营环境未发生重大变化，内部生产秩序及相关经营情况一切正常。二、经向公司控股股东核实：截至目前，无减持公司股份的计划。\\n 关注同花顺财经（ths518），获取更多机会"} }